# Proper-subfield x bases and large-prime rank

**Derived conditional lemma, 2026-10-17 UTC.** Let E descend to Fq, where q=2^d and d is a proper divisor of n. For a point P with x(P) in Fq, q-Frobenius fixes x(P), so Frobenius(P)=+P or −P. Applying its characteristic polynomial gives (q+1−t)P=O or (q+1+t)P=O, where $|t| \le 2\sqrt{q}$. Thus any prime factor of the order of P is at most $q+1+\lfloor 2\sqrt{q} \rfloor$.

## Proper-subfield x bases on descended binary curves

$$x(P) \in F_q \rightarrow \text{Frobenius}_q(P) = \pm P \rightarrow \text{order}(P) \leqslant q + 1 + \lfloor 2\sqrt{q} \rfloor$$

**n=50; largest proper d=25**

maximum point-order ceiling: 33,566,018 < large-prime threshold: 562,949,953,421,312

**n=53; largest proper d=1**

maximum point-order ceiling: 5 < large-prime threshold: 4,503,599,627,370,496

**n=63; largest proper d=21**

maximum point-order ceiling: 2,100,049 < large-prime threshold: 4,611,686,018,427,387,904

Conditional on E descending to Fq and a prime target subgroup r ≥ 2^(n−1). No ordinary-binary non-descended claim.

| Ambient n | Largest proper d | Maximum point-order ceiling |
|---|---|---|
| 50 | 25 | 33,566,018 |
| 53 | 1 | 5 |
| 63 | 21 | 2,100,049 |

The diagram compares these exact ceilings with the explicit condition r≥2^(n−1). The reviewed n=53 Koblitz comparator has r=21,044,858,204,113; this is below that illustrative threshold but above its actual ceiling 5. A pure subfield x base therefore has zero projected rank in that target group.

## Scope and next test

This proof requires E to descend to the selected subfield. It does not cover non-descended ordinary-binary curves, near-subfield sets, displaced sparse predicates, or any complete-cost advantage. The next gate is a target-blind base with independently verified nonzero projected points and rank, then matched-size solver and complete ECDLP comparisons. The canonical frontier graph is unchanged.

Proof and provenance: companion Markdown report; exact integers: research/icperf_subfield_rank_gate_20261017/hasse_bounds.json.